

PROJECT PROPOSAL AND CHARTER

Written by	Reviewed by	Approved by	Latest version & date
Lê Thanh Hoàng Minh Nguyễn Lâm Khải Nguyễn Minh Khang Nguyễn Minh Nhật			1.0 (12/6/2026)
Relevant documents			
https://www.mahjongpictureguide.com/page/3/#tiles : Tiles group images source  Aotenjo: Infinite Hands Official Launch Trailer https://en.wikipedia.org/wiki/Roguelike_deck-building_game			

Version History		
Version	Changelog	Review date
1.0	<i>Initial Documents</i>	
1.1	<i>Clarified scoring model vs. standard Mahjong rules; split technical and gameplay goals with measurable success criteria; specified demo scope (8 antes, ~10 artifacts); added gameplay loop diagram; filled in all milestone dates</i>	

TEAM NAME: SanRokuNana

PROJECT NAME: Malajong

1. CONTACT POINTS

Project Members

Lê Thanh Hoàng Minh

Nguyễn Lâm Khải

Nguyễn Minh Khang

Nguyễn Minh Nhật

2. PROJECT OVERVIEW

Project Description

Project Malajong is a single-player game that takes the tile-matching fun of Mahjong and combines it with the run-based progression of a roguelike deckbuilder. Players will shuffle tiles, utilize passive items (called Artifact), and manipulate hand-scoring algorithms to defeat increasingly point quotas.

Additional Content:

Traditional Mahjong rules decide the winner when one player has met the winning condition of creating 4 combos of three and a pair. Since the game is PvE, the winning condition will be getting enough points by creating combos (combos will grant points according to the difficulty to create)

Combos type example:

Chow - 3 tiles incrementing ex: 123, 345, 678

Pong - 3 tiles of the same ex: 333, 666, 777

Kong - 4 tiles of the same ex: 3333, 5555, 7777

Project Goals

Technical goals

Create a Mahjong hand-evaluation and scoring engine capable of validating Pong/Chow/Kong/pair sets and calculating cumulative chip × multiplier scores in real time, with a target evaluation time under 50ms per play so there is no UI lag.

Design and implement a roguelike progression loop (randomized shop offers, dynamic difficulty scaling across antes, seeded runs) supporting at least 3 distinct seeds for QA.

Implement the codebase using industry-standard software architecture, the MVC pattern for game state/UI separation, and ScriptableObjects for tile, Artifact, and blind data, so designers can add new content without touching core code.

Achieve a stable 60 FPS on a mid-range machine at demo time

Gameplay goals

Deliver a complete run consisting of 8 antes (escalating point quotas), each ending in a Blind, with every 3rd ante culminating in a Boss Wind that introduces a special constraint rule.

Have approximately 10 unique Artifacts at demo stage, each producing a meaningfully different scoring strategy, to demonstrate build diversity and replayability.

Target an average single-run playtime of 15–30 minutes for the demo build, balanced so that a competent player can realistically clear all 8 antes without the run feeling either trivial or unwinnable.

Reach a target of at least 3 distinct viable strategies (e.g. Pong-stacking, Chow-flush, pair-rush) confirmed through internal playtesting, to validate replayability before wider testing.

Project Deliverables

The playable demo will scope to a complete run: 8 antes (including Boss Winds), with approximately 10 purchasable Artifacts available across shop visits. Concretely, the team will deliver:

A playable demo showing the full core game loop: draw → play → score → shop → next round, spanning all 8 antes.

Approximately 10 implemented Artifacts demonstrating distinct scoring synergies.

A documented Git repository containing C# source code and Unity project files.

Additional Content

- Gameplay duration: ~15-30minutes
- Artifacts: artifacts game total (max carry per round: 5)
- Number of rounds to be delivered: 1 tutorial on winning conditions, 2 rounds to introduce game special mechanics (rare combos, special tiles, artifacts)

Scoring System

To avoid conflict between Malajong's requirements and the rules of traditional Mahjong, the scoring system is a custom, roguelike-balanced model inspired by Aotenjo: Infinite Hands, rather than standard fan/han scoring used in real-world Mahjong variants (e.g. Riichi, Hong Kong, MCR).

3. PROJECT SCOPE

In-scope requirements

The core systems involve the development of a 13-tile hand management system including drawing and discarding tiles. The scoring engine will feature an algorithmic calculator for base Mahjong hands combined with limitless multipliers. For economy and progression, a currency system will be built for purchasing tiles and acquiring passive modifiers called Artifact between rounds. The art and audio scope includes the implementation of 2D pixel art assets, basic UI overlays, and integrated sound effects.

Out of scope

This project will not include multiplayer functionality for local or online networking. It will also exclude cross-platform porting to mobile or console devices.

4. TARGET AUDIENCE/USERS

The primary audience consists of fans of deckbuilders and roguelike titles who enjoy mathematical optimization, and theory-crafting. The secondary audience includes traditional board games and Mahjong players seeking an arcade interpretation of the classic game.

5. TECHNICAL APPROACH

Tools and Technologies

We will use the Unity Game Engine for handling complex 2D systems, game logic, and building the project.

C# will be the programming language used to script all game logic, UI interactions, and complex Mahjong hand validations.

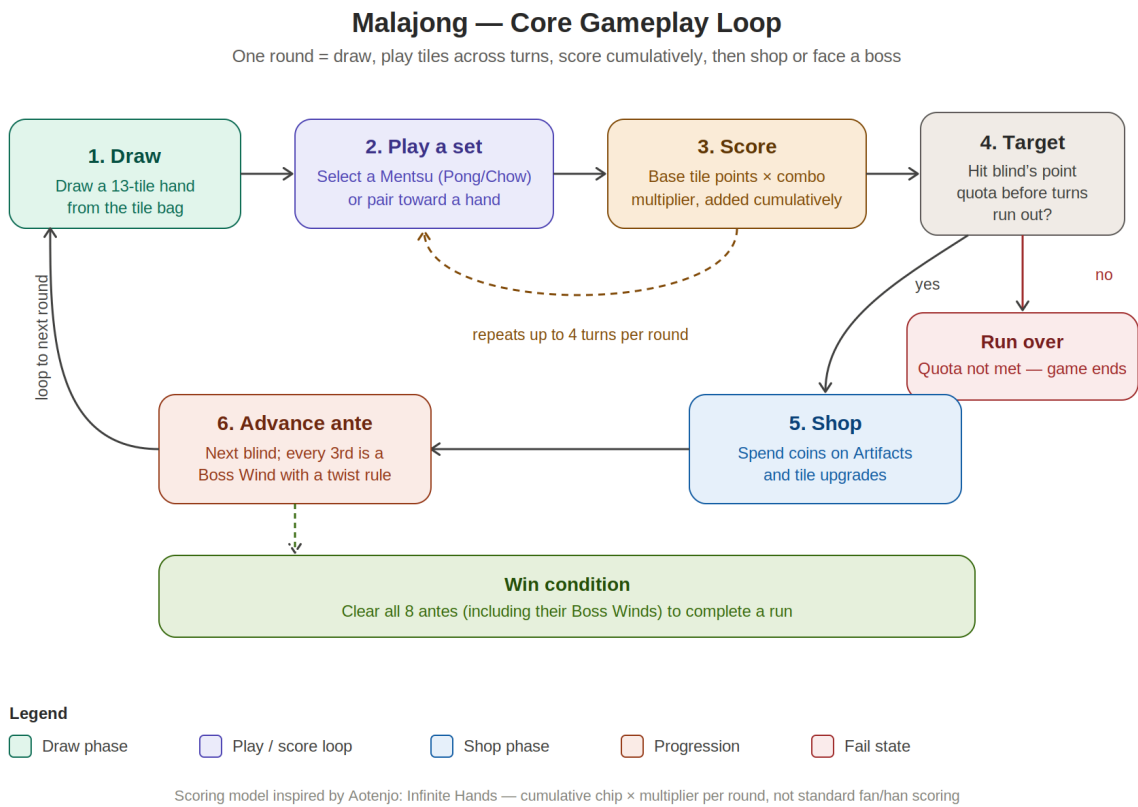
For graphic design, we will use Aseprite, which is the industry standard for creating 2D pixel art, to draw tiles, characters, and UI elements.

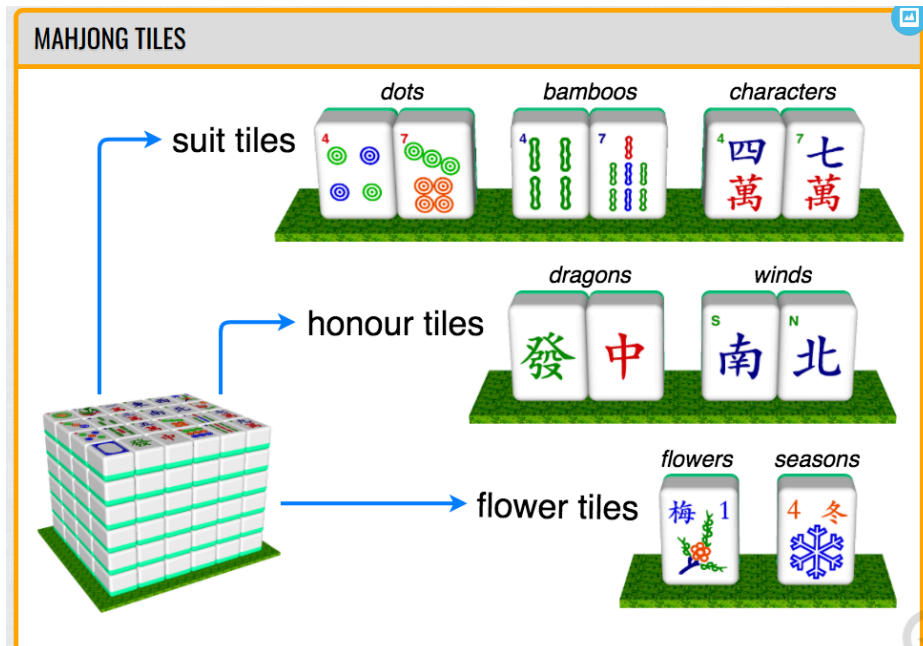
Git and GitHub will be utilized for collaborative code management and version tracking

Jira will be used for project management.
Jira is used to assign tasks and track the progression of work.

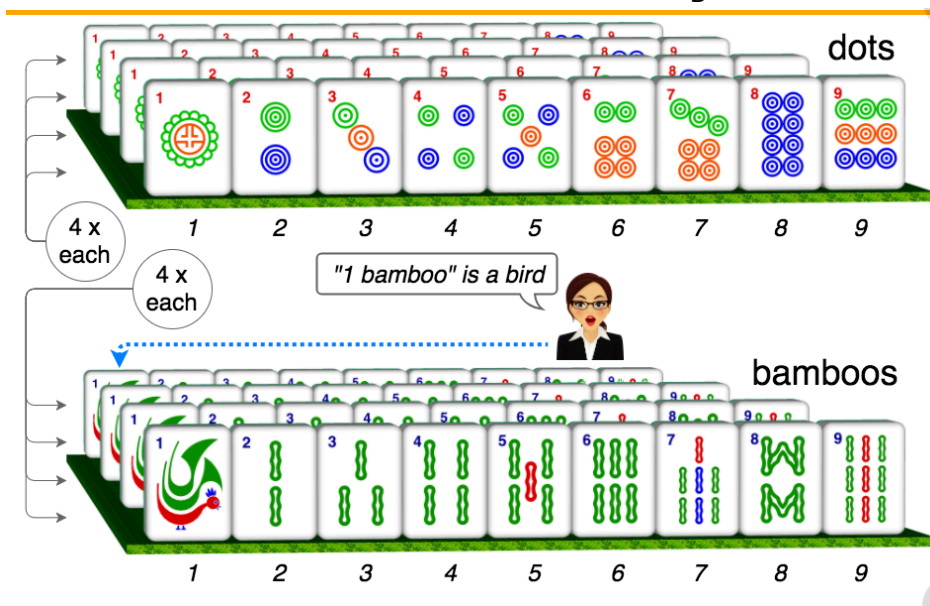
Initial Project Design
Gameplay loop diagram

The diagram below visualizes the full player flow for a single round: draw → play → score → shop → next round, including the win condition and fail state.



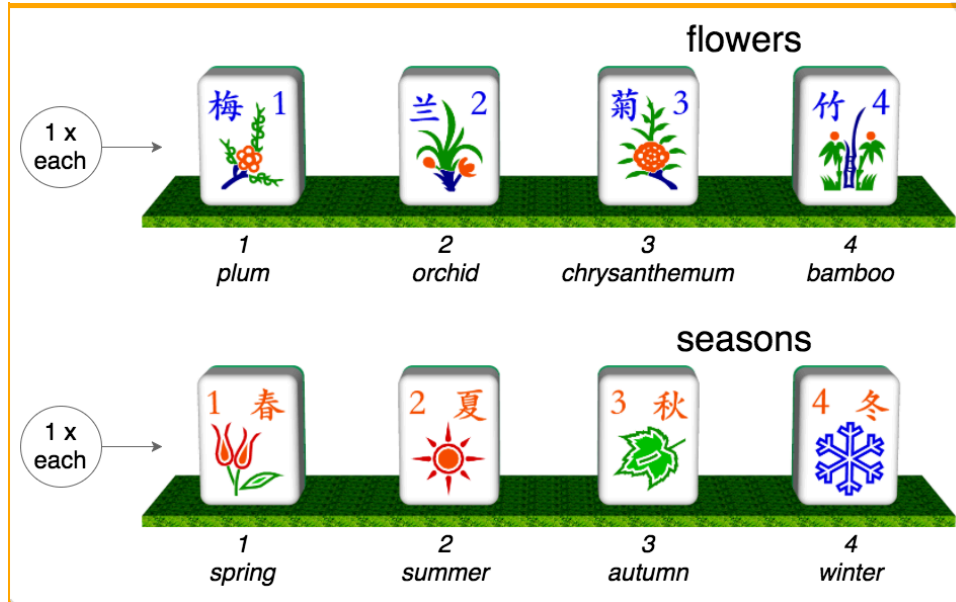


Each group will have 4 tiles for each type, for example dots group will range from 1 to 9 tiles, and each tile will have 4x each, like the image below:



! except for the Flower tiles will only have one for 4 types of tiles:

Flower tiles will be reserved to be a bonus tiles because of their special properties in game



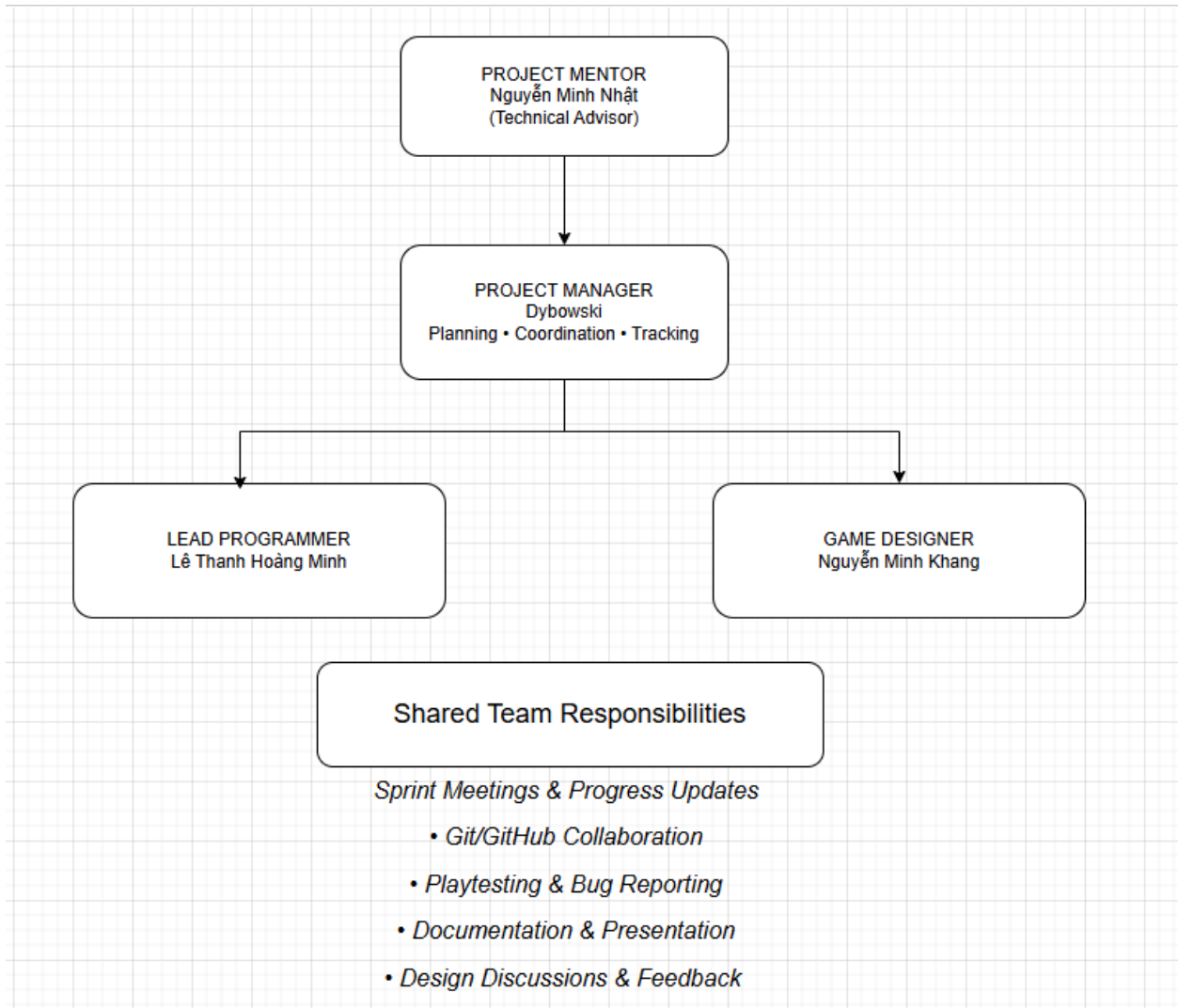
The combo (sets) will be inspired by the original Mahjong (Pong , Chow) and passive items (Artifacts) will be used to increase the multipliers in order to compete with the increasing score required each round.

6. PROJECT MILESTONES

Milestone	Deliverables	Expected Date
Finalize the proposal	Approved proposal & charter (v1.1)	12/6/2026
Jira and Github Repo Setup	Project board, repo, MVC skeleton	15/6/2026
Mahjong Scoring Engine Logic	Working Pong/Chow/pair validator and cumulative chip × multiplier calculator	
Roguelike Loop & Shop Integration	8-ante run loop, shop UI, currency system, Boss Wind rules	
Artifacts System	~10 Artifacts implemented with distinct synergies	
Aseprite Art & UI	Final tile art, UI overlays, sound effects integrated	
QA Testing, Balancing	Internal playtesting across 3 seeds, bug testing, balance pass	18/9/2026
Demo, Presentation	Final playable build and presentation deck	Demo day hehe

7. PROJECT CHARTER

Overall Diagram



Roles and Responsibilities

Role	PIC	Responsibilities	Shared responsibilities
Project Manager	Nguyễn Lâm Khải	Project planning, task associate, balancing, documentation and presentation. Ensure a comfortable working atmosphere.	- Attend weekly team meetings and contribute to discussions. - Communicate progress, blockers, and updates promptly.

Lead Developer, Designer	Lê Thanh Hoàng Minh	Graphing out the initial core mechanic, the gameplay loop, along with with artifacts items,	- Participate in playtesting and provide feedback. - Help maintain project documentation and organization. - Support teammates when needed and collaborate on problem-solving. - Follow coding/design standards and meet assigned deadlines.
Designer, developer	Nguyễn Minh Khang	Design the items and artifacts	
Mentor	Nguyễn Minh Nhật	Provide technical guidance, best practices. Share experience in workflow and project management.	